

Semantic Interpretation and Similarity using TF-IDF and ESA

Step-by-Step process

1. Problem Definition

2. Vocabulary Construction

3. Term Frequency (TF Matrix)

4. Document Frequency and IDF Calculation

Document frequency (df)

IDF Formula

IDF values

5. TF-IDF Matrix

6. ESA vector representation

ESA vector for "green crocodile"

ESA vector for "big crocodile"

ESA vector for "scary crocodile"

7. Cosine Similarity Computation

8. Final answer

Document Collection :

d_1 : "Scary green crocodile."

d_2 : "Scary green big."

d_3 : "Small crocodile."

① Find Unique Words (Vocabulary)

Arrange words alphabetically.

Vocabulary = (big, crocodile, green, scary, small)

② Calculate Term Frequency (TF) matrix.

Find how many times each unique word occurs in a document

Term Doc ↓	big	Crocodile	green	scary	small
d_1	0	1	1	1	0
d_2	1	0	1	1	0
d_3	0	1	0	0	1

③ Calculate document frequency (df) and IDF :

df = Find how many times each unique word occurs in full document collection.

big = 1, crocodile = 2, green = 2,
scary = 2 and small = 1.

Calculate Inverse Document Frequency (IDF)

Calculate inverse document frequency (IDF), which measures how important a word is using the formula:-

$$\text{idf}(\text{term}) = \log \frac{N}{\text{df}(\text{term})}$$

where, $N = 3$, number of documents.

For "big" :- $\text{df}["\text{big}"] = 1$

$$\text{idf}["\text{big}"] = \log\left(\frac{3}{1}\right) = 0.477$$

For "crocodile" :- $\text{df}["\text{crocodile}"] = 2$

$$\text{idf}["\text{crocodile}"] = \log\left(\frac{3}{2}\right) = 0.176$$

For "green" :- $\text{df}["\text{green}"] = 2$

$$\text{idf}["\text{green}"] = \log\left(\frac{3}{2}\right) = 0.176$$

For "scary" :- $\text{df}["\text{scary}"] = 2$

$$\text{idf}["\text{scary}"] = \log\left(\frac{3}{2}\right) = 0.176$$

For "small" :- $\text{df}["\text{small}"] = 1$

$$\text{idf}["\text{small}"] = \log\left(\frac{3}{1}\right) = 0.477$$

Derived df and idf matrix

Term	df	idf
big	1	0.477
crocodile	2	0.176
green	2	0.176
scary	2	0.176
small	1	0.477

④ Calculate TF-IDF (Term-concept) matrix :-

$$tfidf(i, j) = tf(i, j) \times idf(i)$$

where,

$tf(i, j)$ = frequency of term t_i in document d_j

$idf(i)$ = inverted document frequency of term i .

$$tfidf = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} .477 \\ .176 \\ .176 \\ .176 \\ .477 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \times .477 & 1 \times .176 & 1 \times .176 & 1 \times .176 & 0 \times .477 \\ 1 \times .477 & 0 \times .176 & 1 \times .176 & 1 \times .176 & 0 \times .477 \\ 0 \times .477 & 1 \times .176 & 0 \times .176 & 0 \times .176 & 1 \times .477 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & .176 & .176 & .176 & 0 \\ .477 & 0 & .176 & .176 & 0 \\ 0 & .176 & 0 & 0 & .477 \end{bmatrix}$$

Question 1 :-

Calculate semantic interpreter for text

"green crocodile" :-

① Calculate term-frequency (tf) for "green crocodile"

Term	TF	TFIDFQ
big	0	0
crocodile	1	.176
green	1	.176
scary	0	0
small	0	0

② Now, calculate term-concept vector using

$$V_q = \text{tfidf} \times \text{TFIDFQ}$$

$$= \begin{bmatrix} 0 & .176 & .176 & .176 & 0 \\ .477 & 0 & .176 & .176 & 0 \\ 0 & .176 & 0 & 0 & .477 \end{bmatrix} \times \begin{bmatrix} 0 \\ .176 \\ .176 \\ 0 \\ 0 \end{bmatrix}$$

For document d_1 :

$$0 \times 0 + .176 \times .176 + .176 \times .176 + .176 \times 0 + 0 \times 0$$

$$= 0.061952$$

For document d_2 :-

$$.477 \times 0 + 0 \times .176 + .176 \times .176 + .176 \times 0 + 0 \times 0$$

$$= 0.030976$$

For document d_3 :-

$$0 \times 0 + .176 \times .176 + 0 \times .176 + 0 \times 0 + .477 \times 0$$

$$= 0.030976 \quad \therefore V_q = \begin{bmatrix} .062 \\ .031 \\ .031 \end{bmatrix}$$

Final Interpretation :-

document	Score	meaning
d1	.062	most related
d2	.031	less related
d3	.031	less related

Question - 02

Calculate similarity between "big crocodile" and "scary crocodile" based on ESA score.

Let $A = \text{"big crocodile"}$

$B = \text{"scary crocodile"}$

For A :-

Term	TF	TF IDF - A
big	1	.477
crocodile	1	.176
green	0	0
scary	0	0
small	0	0

Now, calculate term-concept vector $V_A = \text{tfidf} \times \text{TFIDFA}$

$$\begin{bmatrix} 0 & .176 & .176 & .176 & 0 \\ .477 & 0 & .176 & .176 & 0 \\ 0 & .176 & 0 & 0 & .477 \end{bmatrix} \times \begin{bmatrix} .477 \\ .176 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore V_A = \begin{bmatrix} .031 \\ 0.228 \\ 0.031 \end{bmatrix}$$

For B :-

Term	TF	TFIDF - B
big	0	0
crocodile	1	.176
green	0	0
scary	1	.176
small	0	0

calculate

$$V_A = \text{tfidf} \times \text{TFIDF - B}$$

$$V_B = \begin{bmatrix} 0 & .176 & .176 & .176 & 0 \\ .477 & 0 & .176 & .176 & 0 \\ 0 & .176 & 0 & 0 & .477 \end{bmatrix} \times \begin{bmatrix} 0 \\ .176 \\ 0 \\ .176 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.062 \\ 0.031 \\ 0.031 \end{bmatrix}$$

Next, Calculate the cosine similarity :-

$$\text{Cosine-similarity}(A, B) = \frac{V_A \cdot V_B}{\|V_A\| \|V_B\|}$$

$$\text{So, } V_A \cdot V_B = \begin{bmatrix} .031 \\ .228 \\ .031 \end{bmatrix} \cdot \begin{bmatrix} .062 \\ .031 \\ .031 \end{bmatrix}$$

$$= .031 \times .062 + .228 \times .031 + .031 \times .031$$

$$= .001922 + .007068 + .000961$$

$$= 0.009951$$

Now,

$$\begin{aligned} \|V_A\| &= \sqrt{.031^2 + .228^2 + .031^2} = \sqrt{.000961 + .051984 + .000961} \\ &= \sqrt{.053906} = 0.23217 \end{aligned}$$

$$\begin{aligned} \|V_B\| &= \sqrt{.062^2 + .031^2 + .031^2} = \sqrt{.003844 + .000961 + .000961} \\ &= \sqrt{.005766} = 0.075934 \end{aligned}$$

$$\begin{aligned} \therefore \text{cosine-similarity}(A, B) &= \frac{0.009951}{.23217 \times .075934} \\ &= \frac{.009951}{.0176296} \\ &= 0.5644676 \end{aligned}$$

Final Answer :-

1. ESA vector for "green crocodile" = $\begin{bmatrix} .062 \\ .031 \\ .031 \end{bmatrix}$

2. Similarity ("big crocodile", "beary crocodile")
= 0.5644.